

ONLINE FOOD DELIVERY ANALYTICS

End-to-End BI & Advanced SQL Performance Case Study

TECH STACK: POSTGRESQL | POWER BI | PYTHON | DAX

Executive Summary

In a highly competitive online food delivery landscape, data-driven optimization is the primary engine for sustainable revenue growth and operational excellence. This comprehensive case study presents a Business Intelligence and Performance Analysis solution inspired by industry leaders like Zomato, Swiggy, and Uber Eats.

By designing a robust PostgreSQL database architecture, engineering automated pipeline logic, and layering an interactive Power BI dashboard, this project translates transactional noise into concrete business insights. The end goal is to dissect revenue drivers, refine delivery efficiency, map payment methods, and craft structured data foundations for strategic growth.

\$1.34M

TOTAL REVENUE

1,001

TOTAL ORDERS

296

ACTIVE
CUSTOMERS

40.22m

AVG DELIVERY
TIME

Core Business Insights

- **Geographic Performance:** *Mumbai* is established as the premier market leader, generating the highest sales density across all metrics (\$0.31M), closely pursued by *Chennai* (\$0.29M) and *Kolkata* (\$0.26M).
- **Cuisine Popularity Index:** High consumer density is anchored heavily within *South Indian* and *Indian* cuisines, followed steadily by *Italian*, *Fast Food*, *Chinese*, and *Desserts*.
- **Macroeconomic Warning Signal:** Time-series revenue visualization reveals a consistent **declining month-over-month trend** spanning from March through September, flagging an urgent requirement for targeted counter-seasonal marketing frameworks.
- **Transactional Preferences:** Digital transactions aggressively dominate customer preference, with **UPI (\$460.62K / 34.4%)** and **Cards (\$435.78K / 32.54%)** processing over two-thirds of platform volume.
- **Demographics & Pareto Impact:** Male consumers contribute a marginally higher revenue stream (\$758.83K) compared to female consumers (\$580.19K). Crucially, a small, highly concentrated segment of high-value patrons (Gold tier) generates a massive percentage of overall platform GMV.

⚠ Operational Bottleneck Identified: The platform's overall average delivery duration sits at 40.22 minutes. In the ultra-fast commerce framework, this indicates severe friction points within logistics, kitchen preparation sync, or courier routing.

Strategic Business Recommendations

1. Geo-Targeted Scale & Expansion

Concentrate core capital expenditures and hyper-local brand partnerships within high-yield territories—specifically scaling footprint in Mumbai and Chennai to maximize market dominance.

2. Reversing the Churn Trend via Micro-Campaigns

Deploy aggressive seasonal and festive promotional campaigns during the historic low-performing period (March to September). Leverage personalized, high-margin cuisine discount bundles (South Indian & Indian) to stimulate user order frequency.

3. Logistics Refinement & Delivery Incentives

Introduce predictive geo-fencing for delivery agents, automate kitchen dispatch times, and roll out performance-tied cash incentives to drive the average delivery baseline down toward a target of under 30 minutes.

4. Retention Framework for Power Users

Establish structured loyalty programs targeting the top-tier customer segment identified via SQL window functions. Prevent user churn through exclusive perks, zero delivery fees, and priority customer service.

Technical Implementation Architecture

The backend engine utilizes a PostgreSQL architecture structured across 5 core relation schemas: Customers, Restaurant, Orders, Order_Items, and Delivery_Agents. Advanced analytics queries were implemented to drive automation and speed.

1. Advanced Data Engineering & Automation Logic

To preserve data integrity at the database layer and prevent bad records from downstream processing, custom PL/pgSQL database objects and triggers were deployed:

```
-- Automated Trigger: Prevent negative discount entries
CREATE OR REPLACE FUNCTION online_p.prevent_negative_discount()
RETURNS TRIGGER AS $$
BEGIN
    IF NEW.discount < 0 THEN
        RAISE EXCEPTION 'Discount value cannot be negative';
    END IF;
    RETURN NEW;
END;
$$ LANGUAGE plpgsql;

CREATE TRIGGER trg_prevent_negative_discount
BEFORE INSERT OR UPDATE ON online_p.orders
FOR EACH ROW EXECUTE FUNCTION online_p.prevent_negative_discount();
```

2. Dynamic Customer Segmentation (RFM / Value Tiers)

Utilizing Common Table Expressions (CTEs) and conditional windowing logic, customers are partitioned into actionable promotional cohorts directly from transaction history:

```
WITH customer_spending AS (
    SELECT c.customer_id, c.name, ROUND(SUM(o.order_amount)::numeric, 2) AS total_spent
    FROM online_p.orders o
    JOIN online_p.customers c ON o.customer_id = c.customer_id
    GROUP BY c.customer_id, c.name
)
SELECT customer_id, name, total_spent,
CASE
    WHEN total_spent >= 1500 THEN 'Gold Tier'
    WHEN total_spent >= 500 THEN 'Silver Tier'
    ELSE 'Bronze Tier'
END AS customer_category
FROM customer_spending
ORDER BY total_spent DESC;
```

Future Scope & Predictive Roadmaps

To further scale the platform's technological capabilities, upcoming operational phases will focus on migrating static analytical metrics into predictive machine learning modules:

- **Customer Churn Prediction:** Deploying Scikit-learn random forest algorithms to preemptively highlight accounts indicating drop-offs in order cadence.
- **Personalized Content Engines:** Utilizing collaborative filtering mechanisms to display real-time curated cuisine feeds based on historic contextual data.

- **Demand & Surge Forecasting:** Building predictive LSTM time-series networks to dynamically align delivery driver availability with projected order spikes by city.